Lectures 7-8

Bivariate random variables: Joint distributions, marginal distributions and conditional distributions

Definition. Let X and Y be two r.v.'s. The joint or bivariate distribution of X and Y is the collection of all probabilities of the form:

$$\mathbb{P}(\{(s,t) \in S \times S : (X(s), Y(t)) \in C\})$$

Notation $P(\{(X,Y) \in C\})$, for all sets $C \subset \mathbb{R}^2$ such that $\{(X,Y) \in C\}$ is an event.

Note The definition of "event" was given at pages 11-12 in the first handout.

Discrete Joint Distributions

Definition 1 Let X and Y be random variables. if the ordered pair (X,Y) has only at most countably possible values (x,y), then (X,Y) have a joint discrete distribution.

Definition 2 The joint probability function of (X,Y) is the function f defined on $\mathbb{R}^2$ and given by:

$$f(x,y) = \mathbb{P}(X=x,\ Y=y).$$

Theorem Let (X,Y) have a discrete joint distribution f. Then:

1. $f(x,y)=0$, if (x,y) is NOT a possible value for (X,Y)
2. $\sum_{\text{All }(x,y)} f(x,y) = 1$
3. $\mathbb{P}[(X,Y)\in C] = \sum_{(x,y)\in C} f(x,y)$.

Example X = # of cars owned by a household.

Y = # of television sets owned by a household.

- Assume: $X \in \{1, 2, 3\}$; $Y \in \{1, 2, 3, 4\}$.
- Assume further that the joint p.f. of (X, Y) is as given below:

x \ y	1	2	3	4	Total.
1	0.1	0	0.1	0	0.2
2	0.3	0	0.1	0.2	0.6
3	0	0.2	0	0	0.2
Total	0.4	0.2	0.2	0.2	

- Calculate (a) $\mathbb{P}(X \geq 2, Y \geq 2)$

"And"

(b) $\mathbb{P}(X = 1)$.

(a) $\mathbb{P}(X \geq 2, Y \geq 2) = f(2,2) + f(2,3) + f(2,4)$
$+ f(3,4) + f(3,2) + f(3,3)$

$= 0.5$

(b) $\mathbb{P}(X = 1) = \sum_{y=1}^{4} f(1,y) = 0.2$.

Continuous Joint Distributions
(Joint p.d.f)

Definition 1 X and Y have a **continuous joint distribution** if there exists a function $f \geq 0$, f defined on $\mathbb{R}^2$, such that for any $C \subseteq \mathbb{R}^2$ we have

$$\mathbb{P}((X,Y) \in C) = \iint_C f(x,y)\,dx\,dy.$$

Theorem In order to call $f: \mathbb{R}^2 \to \mathbb{R}$ a joint p.d.f., f must satisfy:

① $f(x,y) \geq 0$, for all $x \in \mathbb{R}^2$

② $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1$.

Important results

Let (X,Y) have a continuous joint p.d.f. on $\mathbb{R}^2$. Then:

① $\mathbb{P}((X,Y)=(x,y)) = 0$;

$\mathbb{P}((X,Y) \in \bigcup_{i=1}^{\infty}\{(x_i,y_i)\}) = 0$.

② Consider sets of the following form:

$A =: \{(x,y) \in \mathbb{R}^2 \mid y = g(x),\ a < x < b\}$,

$B =: \{(x,y) \in \mathbb{R}^2 \mid x = h(y),\ c < y < d\}$,

where g and h are continuous functions.

Then $\mathbb{P}((X,Y) \in A) = \mathbb{P}((X,Y) \in B) = 0$.

Example 1 Let

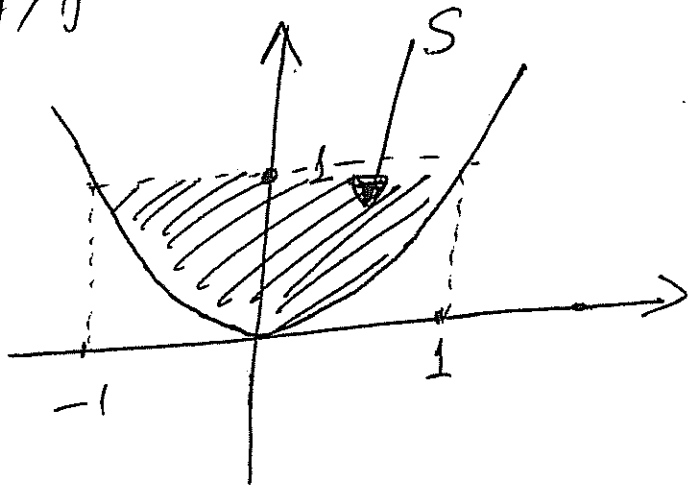
$$f(x,y) = \begin{cases} c x^2 y, & \text{for } x^2 \le y \le 1 \\ 0, & \text{otherwise.} \end{cases}$$

(1) Find c for which $f(x,y)$ qualifies as a pdf of some (X,Y).

(2) For c in (1), calculate $\mathbb{P}(X \ge Y)$.

Solution. (1) • First, find the support of f, that is find $S = \{(x,y) \in \mathbb{R}^2 \mid f(x,y) > 0\}$.

Clearly, for us. $S = \{(x,y) \in \mathbb{R}^2 \mid x^2 \le y \le 1\}$.



- Next find $c>0$ s.t. $\iint_{\mathbb{R}^2} f(x,y)\,dx\,dy = 1$.

For us, this simplifies to:

$$1 = \iint_S f(x,y)\,dx\,dy$$

$$= \int_{-1}^{1} \int_{x^2}^{1} c \cdot x^2 y \, dy \, dx$$

$$= c \cdot \int_{-1}^{1} x^2 \left[\frac{y^2}{2} \Big|_{x^2}^{1} \right] dx$$

$$= c \cdot \int_{-1}^{1} x^2 \cdot \left[\frac{1}{2} - \frac{x^4}{2} \right] dx$$

$$= c \left\{ \frac{1}{2} \int_{-1}^{1} x^2 dx - \frac{1}{2} \int_{-1}^{1} x^6 dx \right\}$$

$$= \frac{4}{21} c \implies \boxed{c = \frac{21}{4}}$$

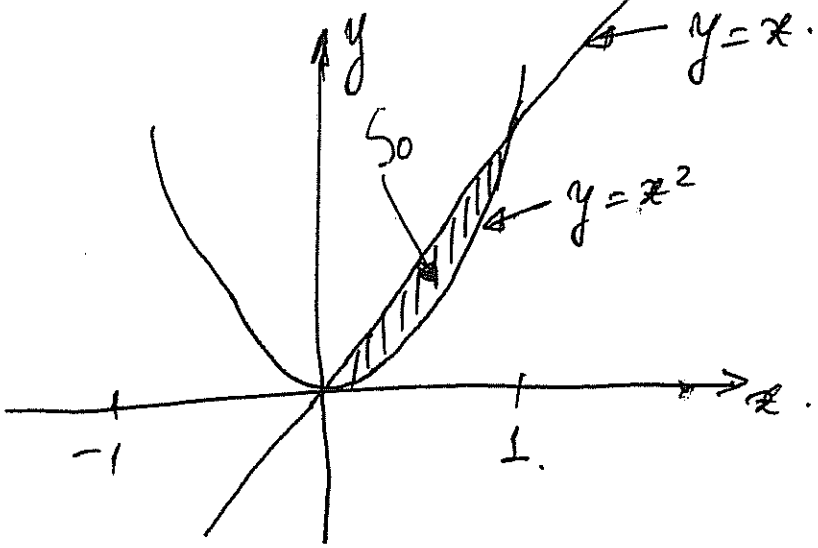
(b) $\mathbb{P}(X \geq Y) = \iint_{\{(x,y)\in\mathbb{R}^2 : x \geq y\}} f(x,y)\,dx\,dy$

definition of joint distributions

continuous

$$= \iint_{\{(x,y)\in\mathbb{R}^2 \,/\, x \geq y\} \cap S} \frac{21}{4} x^2 \cdot y \, dx\,dy \,.$$

Let S_0 denote the set intersection above. S_0 is drawn below:



Therefore: $S_0 = \{(x,y) \mid 0 \leq x \leq 1,\ x^2 \leq y \leq x\}$ and

$S_0 \longrightarrow$ ctud over

$$\mathbb{P}(X \geq Y) = \int_0^1 \frac{21}{4} x \left[\int_{x^2}^{x} y^2 \, dy \right] dx$$

$$= \frac{21}{4} \int_0^1 x \left(\frac{y^3}{3} \Big|_{x^2}^{x} \right) dx$$

$$= \frac{21}{4} \int_0^1 x \left[\frac{x^3}{3} - \frac{x^6}{3} \right] dx$$

$$= \frac{21}{4} \int_0^1 \left[\frac{x^4}{3} - \frac{x^7}{3} \right] dx$$

$$= \frac{3}{20} .$$

Remarks ① It is <u>not true</u> that if X and Y both have cont. p.d.f's the joint is also a continuous p.d.f.

Example Let X be continuous and define $Y = X$, therefore Y is also continuous. Is the joint distribution of (X, X) continuous?

NO: • If it were continuous, then by "Important result ②" at page 34 we would need to have $\mathbb{P}(X = X) = 0$.

• Obviously, for any r.v. X, $\mathbb{P}(X = X) = 1$.

Therefore (X, X), with $Y = X$ <u>cannot</u> have a jointly continuous distribution.

② If X is discrete and Y is continuous we obtained what is called a "mixed bivariate distribution

<u>Reading exercise</u>: @Subsection "Mixed bivariate distributions", pages 123 – 125. <u>NOT</u> required for exams.

Bivariate Cumulative distribution functions

- $F(x,y) =: \mathbb{P}(X \le x,\ Y \le y)$.
- If (X,Y) have a <u>continuous</u> joint p.d.f. f then:

$$F(x,y) = \int_{-\infty}^{x}\int_{-\infty}^{y} f(r,s)\,dr\,ds.$$

- $$f(x,y) = \frac{\partial^2 F(x,y)}{\partial x\,\partial y} = \frac{\partial^2 F(x,y)}{\partial y\,\partial x}.$$

Reading exercises: Examples 3.4.14, 3.4.15 pages 127–128.

- <u>Note</u> For <u>any</u> (X,Y):

$$P(a < X \le b \text{ and } c < Y \le d) = F(b,d) - F(a,d) - F(b,c) + F(a,c).$$

<u>If</u> (X,Y) is <u>continuous</u>, you don't have to pay attention to whether the intervals are closed or open.

• Let $F_1(x)$ be the cdf of X,
$F_2(y)$ be the cdf of Y.

Result (Th. 3.4.5 page 126)

$$\begin{cases} F_1(x) = \lim_{y \to \infty} F(x,y), \\ \text{and} \\ F_2(y) = \lim_{x \to \infty} F(x,y). \end{cases}$$

• $F_1(x)$ is called the marginal cdf of X.

• $F_2(y)$ is the marginal cdf of Y.

Marginal distributions

- Let $(X, Y) \sim f(x, y)$, a joint p.d.f.
- What are $f_X(x)$ and $f_Y(y)$, the individual p(d)f's of X and Y, respectively?
- f_X is called the marginal p(d)f of X
  f_Y " " " p(d)f of Y.

Note p(d)f → p.f. → for discrete distributions
p(d)f → p.d.f. → for continuous distributions.

Discrete marginals

If $f(x, y)$ is the discrete joint p.f. of (X, Y), then:

$$\begin{cases} f_X(x) = \sum_{\text{All } y} f(x, y) \\ f_Y(y) = \sum_{\text{All } x} f(x, y). \end{cases}$$

- Example 3.5.2 pages 131/132 → And Example at page (32) notes, above.

$$X \begin{pmatrix} 1 & 2 & 3 \\ 0.2 & 0.6 & 0.2 \end{pmatrix} \quad \text{and}$$

$$Y \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0.4 & 0.2 & 0.2 & 0.2 \end{pmatrix}$$

Continuous marginals

If $(X,Y) \sim f(x,y)$, where $f(x,y)$ is a probability density function (p.d.f), then

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)\,dy \quad \left(\begin{array}{l}\text{We "integrate out"} \\ y.\end{array}\right)$$

and

$$f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\,dx \quad \left(\begin{array}{l}\text{We "integrate out"} \\ x.\end{array}\right)$$

Example (Return to Example 1, page 35.)

$$f(x,y) = \begin{cases} \frac{21}{4}x^2 y, & \text{for } x^2 \le y \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find $f_X(x)$ and $f_Y(y)$.

Solution

- $f_X(x) = \int_{-\infty}^{\infty} f(x,y)\,dy$.

$$= \int_{x^2}^{1} \frac{21}{4} \cdot x^2 \cdot y \cdot dy$$

$$= \frac{21}{4} \cdot x^2 \cdot \int_{x^2}^{1} y\,dy$$

$$= \frac{21}{4} \cdot x^2 \cdot \left. \frac{y^2}{2} \right|_{y=x^2}^{y=1} = \frac{21}{4} \cdot x^2 \left(\frac{1}{2} - \frac{x^4}{2} \right)$$

Therefore :

$$f_X(x) = \begin{cases} \frac{21}{8} \cdot x^2(1-x^4), & \text{for } -1 \le x \le 1 \\ 0, & \text{otherwise.} \end{cases}$$

- $$f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\, dx$$

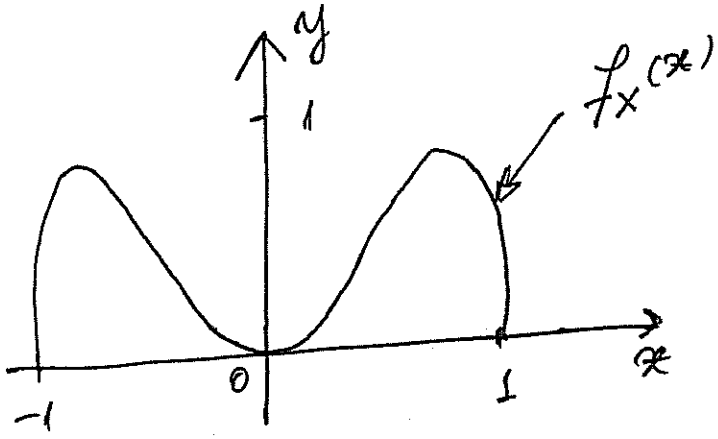
$$= \int_{-\sqrt{y}}^{-\sqrt{y}} \frac{21}{4} x^2 y \, dx$$

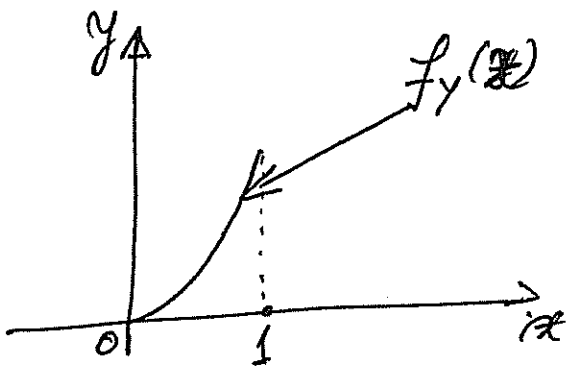
(= by definition of $f(x,y)$)

$$= \frac{21}{4} \cdot y \cdot \left(\frac{x^3}{3} \Big|_{x=-(y)^{1/2}}^{x=(y)^{1/2}} \right)$$

$$= \frac{21}{4} \cdot y \left[\frac{y^{\frac{3}{2}}}{3} + \frac{y^{\frac{3}{2}}}{3} \right] = \frac{21}{4} \cdot y \cdot \frac{2y^{\frac{3}{2}}}{3}$$

$$= \frac{7}{2} y^{\frac{5}{2}}. \quad //$$





$$f_Y(y) = \begin{cases} \frac{7}{2} \cdot y^{\frac{5}{2}}, & \text{for } 0 \le y \le 1 \\ 0, & \text{otherwise.} \end{cases}$$

$\Leftrightarrow$ y is just a dummy variable, we can also write

$$f_Y(x) = \begin{cases} \frac{7}{2} \cdot x^{\frac{5}{2}}, & \text{for } 0 \le x \le 1 \\ 0, & \text{otherwise.} \end{cases}$$

IMPORTANT Given $f(x,y)$ we can find $f_X(x)$ and $f_Y(y)$. The reverse is NOT true:

Given marginals, we cannot recover the joint distribution, without additional info.

Reading (Not required for exams): Th. 3.5.3 and Example 3.5.4.